Snagboot: vendor-agnostic, open-source and developer-friendly recovery and reflashing tool

Thomas Petazzoni, Bootlin

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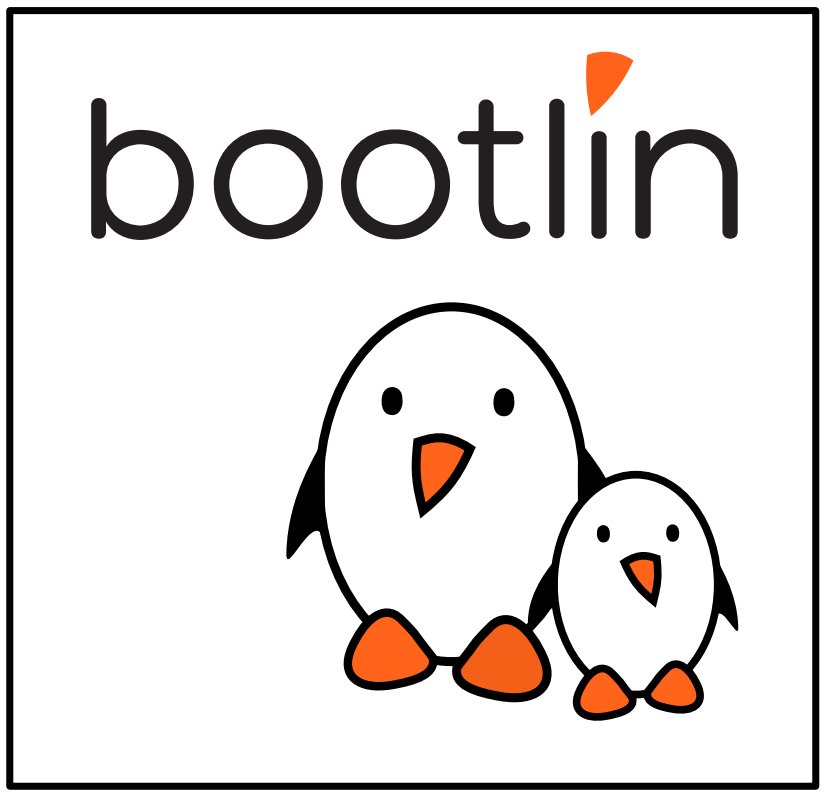
Corrections, suggestions, contributions and translations are welcome!

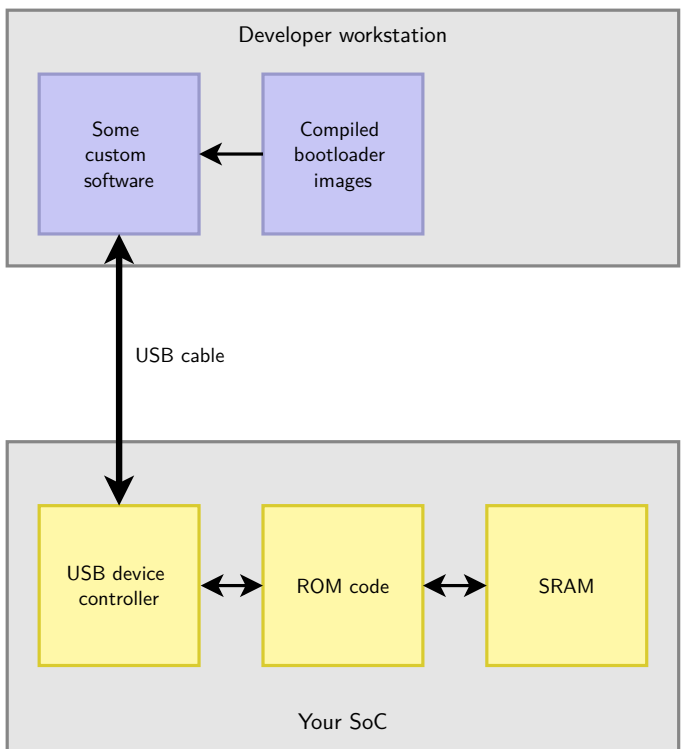


whoami

- Thomas Petazzoni
- CEO of *Bootlin*
  - Engineering company specialized in **Embedded Linux** and **Zephyr**
  - 28 people
  - Engineering services
  - Training services
  - Very strong open-source focus
  - We are **hiring**
- Former contributor to the *Linux kernel*: 900+ patches
- Co-maintainer and contributor to *Buildroot*

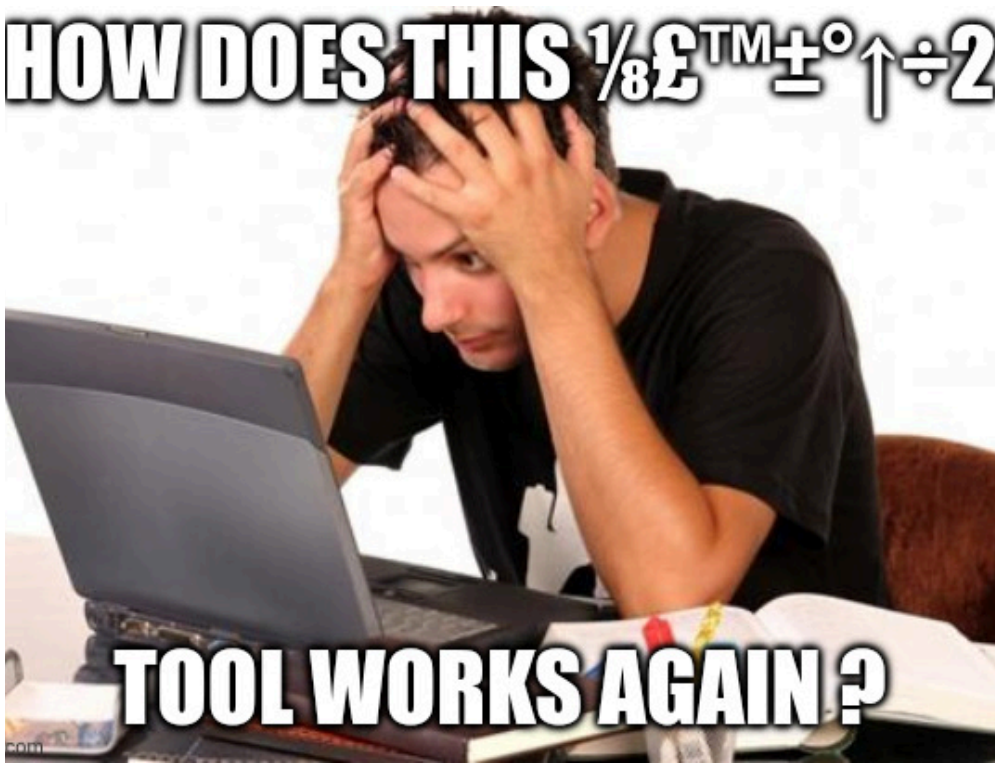
ROM code and recovery

- Virtually all modern embedded SoCs integrate a ROM code
- Responsible for initial boot steps
- Generally provides a **recovery** mechanism
- Useful for
  - Factory flashing
  - Recovering bricked boards
- Typically uses a specific protocol on USB, sometimes UART or Ethernet
- Allows to send a binary payload into internal SRAM, and run it



Previous situation

- Each vendor has its **own** tool
  - NXP: ~~mfgtools~~ → uuu
  - ST: STM32 Cube Programmer
  - Microchip: SAM-BA
  - Rockchip: rkdeveloptool
  - etc.
- One tool per vendor is **annoying**
- Vendor tools are not always open-source
- Vendor tools are not always flexible/customizable



Enter Snagboot

- **One** project to rule them all
- **Vendor-agnostic**, supports **many** platforms
- **Open-source**: GPLv2
- **Hacker-friendly**: fully implemented in Python
- **Three** components
  - snagrecover
  - snagflash
  - snagfactory
- Works on Linux and Windows
- Developed and maintained by Bootlin engineer **Romain Gantois**
- https://github.com/bootlin/snagboot
- https://snagboot.readthedocs.io

- Allwinner sunxi
- STMicroelectronics STM32MP
- Microchip SAMA5
- NXP i.MX6, i.MX7, i.MX8, i.MX9
- Texas Instruments AM335x, AM6x
- Xilinx/AMD Zynq UltraScale+
- Intel Keembay
- Broadcom BCM2711 (RPi 4) and BCM2712 (RPi 5)
- AMLogic: many SoCs supported

snagrecover

- Performs the recovery step: pushing a working bootloader on the target
  - Typically U-Boot
  - Bootloader is then expected to expose a USB gadget such as fastboot, UMS or DFU

```
$ cat fw.yaml
paths-relative-to: THIS_FILE

tiboot3:
  path: tiboot3_evm.bin
tispl:
  path: tispl_evm.bin
u-boot:
  path: u-boot_evm.img
```

```
$ snagrecover -s am625 -f fw.yaml
2026-01-28 17:26:42,257 [INFO] Starting recovery of am625
board
2026-01-28 17:26:42,260 [INFO] Installing firmware tiboot3
[...]
2026-01-28 17:26:44,082 [INFO] Installing firmware tispl
[...]
2026-01-28 17:26:45,648 [INFO] Installing firmware u-boot
[...]
2026-01-28 17:26:52,621 [INFO] Done recovering am625 board
$
```

snagflash

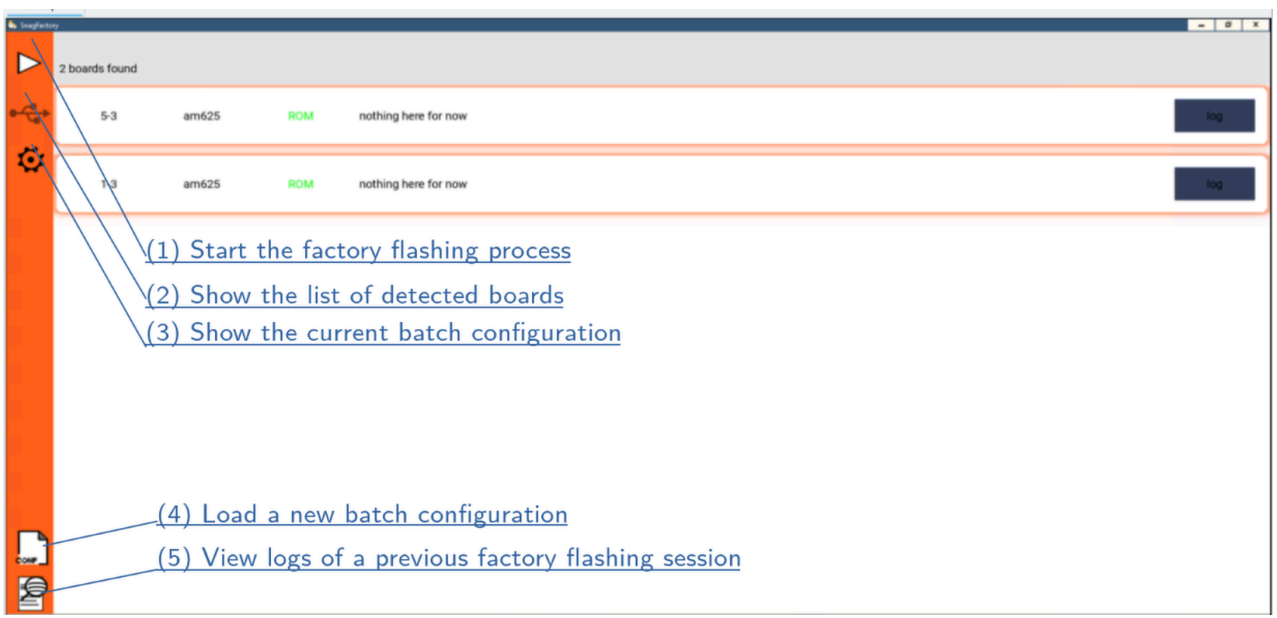
- Performs the flashing step
- Assumes “something” is running on the target
  - Using **snagrecover** or some other mean
  - Typically U-Boot
  - Exposes Fastboot, USB Mass Storage or DFU

```
$ cat flash.cmd
set fb-addr 0xd0000000
set target mmc0

flash "random-1k.bin" 0 hwpart 1
flash "random-1M.bin" 1024
flash "random-100M.bin" 2048
flash "random-1G.bin.bz2" 2048

set target nor0
set eraseblk-size 0x40000

run oem_run:mtd list
flash "random-1k.bin" 0x40000
```

```
$ snagflash -P fastboot-uboot -p 0451:6165 -I flash.cmd
2026-01-28 17:27:04,916 [INFO] Running snagflash using
protocol fastboot-uboot
[...]
2026-01-28 17:27:04,929 [INFO] running command flash
"random-1k.bin" 0 hwpart 1
[...]
2026-01-28 17:27:05,321 [INFO] flashed 4096/? bytes
[...]
2026-01-28 17:29:14,805 [INFO] Flashing file random-1k.bin
2026-01-28 17:29:14,805 [INFO] Flashing to MTD device...
2026-01-28 17:29:15,639 [INFO] fastboot OKAY
2026-01-28 17:29:16,083 [INFO] flashed 4096/? bytes
```

snagfactory

- Factory flashing
- Builds on top of snagrecover/snagflash
- Multiple boards in parallel
- GUI oriented for operators



What to remember ?

UUU, STM32
Cube Programmer,
SAM-BA,
rkdeveloptool,
sunxi-fel, etc.

Snagboot